CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 13: Pythagoras Theorem

Every formula and result of the chapter on one sheet

1	Pythagoras theorem	$AC^2 = AB^2 + BC^2$, with $\angle B = 90^\circ$
2	Hypotenuse	$c = \sqrt{a^2 + b^2}$
3	Other side	$a = \sqrt{c^2 - b^2}$
4	Converse	$c^2 = a^2 + b^2$ gives a right angle opposite c
5	Acute or obtuse	$c^2 < a^2 + b^2$ acute; $c^2 > a^2 + b^2$ obtuse
6	Triplet generator	$(2m, m^2 - 1, m^2 + 1)$
7	Diagonal of a rectangle	$\sqrt{l^2 + b^2}$
8	Diagonal of a square	$a\sqrt{2}$
9	Rhombus	$4 \times \text{side}^2 = d_1^2 + d_2^2$
10	Altitude of an equilateral triangle	$\frac{\sqrt{3}}{2}a$
11	Area of an equilateral triangle	$\frac{\sqrt{3}}{4}a^2$
12	Altitude on the hypotenuse	$\frac{ab}{c}$
13	Perpendicular AD on BC	$AB^2 - BD^2 = AC^2 - CD^2$